

Ryan Hailey

Mechanical Engineering Student - Robotics and Automation

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Technical Skills

Programming & Controls: C, C++, Python, MATLAB, PID Controllers, Automation

CAD, CAE & Manufacturing: SolidWorks, Onshape, Creo, CadQuery, ANSYS Workbench, 3D Printing, Additive Manufacturing, Mechanical Assembly, Prototyping

Optimization: CasADi, IPOPT, AeroSandbox

Robotics & Embedded: ROS2/Nav2, ESP32, Teensy 4.1, Pixhawk, Betaflight, Sensor Fusion, Drivetrain Design, System Integration

Software & Tools: LabVIEW, Simulink, Excel, Microsoft Office, Google Workspace

Projects

Autorotation Recovery Drone (Capstone)

Spring 2026

- Led aerodynamic optimization (CasADi/IPOPT nonlinear programming) of a 3-blade autorotation system to maximize unpowered descent time for a 3D-printed rotorcraft.
- Designed a semi-autonomous flight controller (ESP32-S3 + Betaflight) with Kalman-filtered altitude estimation and cascaded velocity control.
- Automated a CadQuery pipeline to generate parametric blade geometry from optimizer output; design won the competition.

RoboMasters Combat Robot Competition

Fall 2024 - Summer 2026

- Co-founded UTA's inaugural RoboMasters team and served as Drivetrain Lead, owning subsystem design intent, coordinating interfaces and schedules across team leads, and mentoring junior members.
- Designed and manufactured 4 custom drivetrain components using CAD modeling and Bambu 3D printers.
- Performed modal, stiffness, and strength simulations in ANSYS Workbench on the aluminum chassis to evaluate vibration behavior, deflection, and structural integrity.

Autonomous UGV Development

Spring 2026

- Developed a tracked autonomous vehicle progressing from GPS waypoint navigation to encoder-based dead reckoning and LiDAR localization using ROS2/Nav2.
- Integrated a Teensy 4.1, Intel NUC, motor encoders, and RPLIDAR A2; implemented differential-drive control, PID tuning, and a custom ROS2 serial bridge for motor commands.
- Designed and iterated drivetrain components and sensor mounts in SolidWorks, resolving mechanical failures through load-path analysis, print-orientation changes, and reinforced hardware.
- Achieved reliable multi-waypoint navigation using SLAM Toolbox, AMCL localization, and Nav2 path planning and obstacle avoidance.

Education

University of Texas at Arlington, Bachelor of Science in Mechanical Engineering

Aug 2022 – present

- GPA: 3.6
- Pi Tau Sigma Honor Society Member

Professional Experience

Mechanical Engineer CWEP, Lockheed Martin

July 2025 – present

- Supported life-cycle development of complex electromechanical systems for HIMARS and M270 MLRS platforms, balancing performance, schedule, and stakeholder requirements.
- Interpreted customer and program requirements to support design decisions and resource prioritization across engineering teams.
- Coordinated with mechanical, electrical, and manufacturing stakeholders to align system changes with delivery timelines.
- Analyzed mechanical and electrical system issues to assess operational risk and recommend corrective actions.
- Developed 3D models and engineering documentation to communicate design intent across teams.